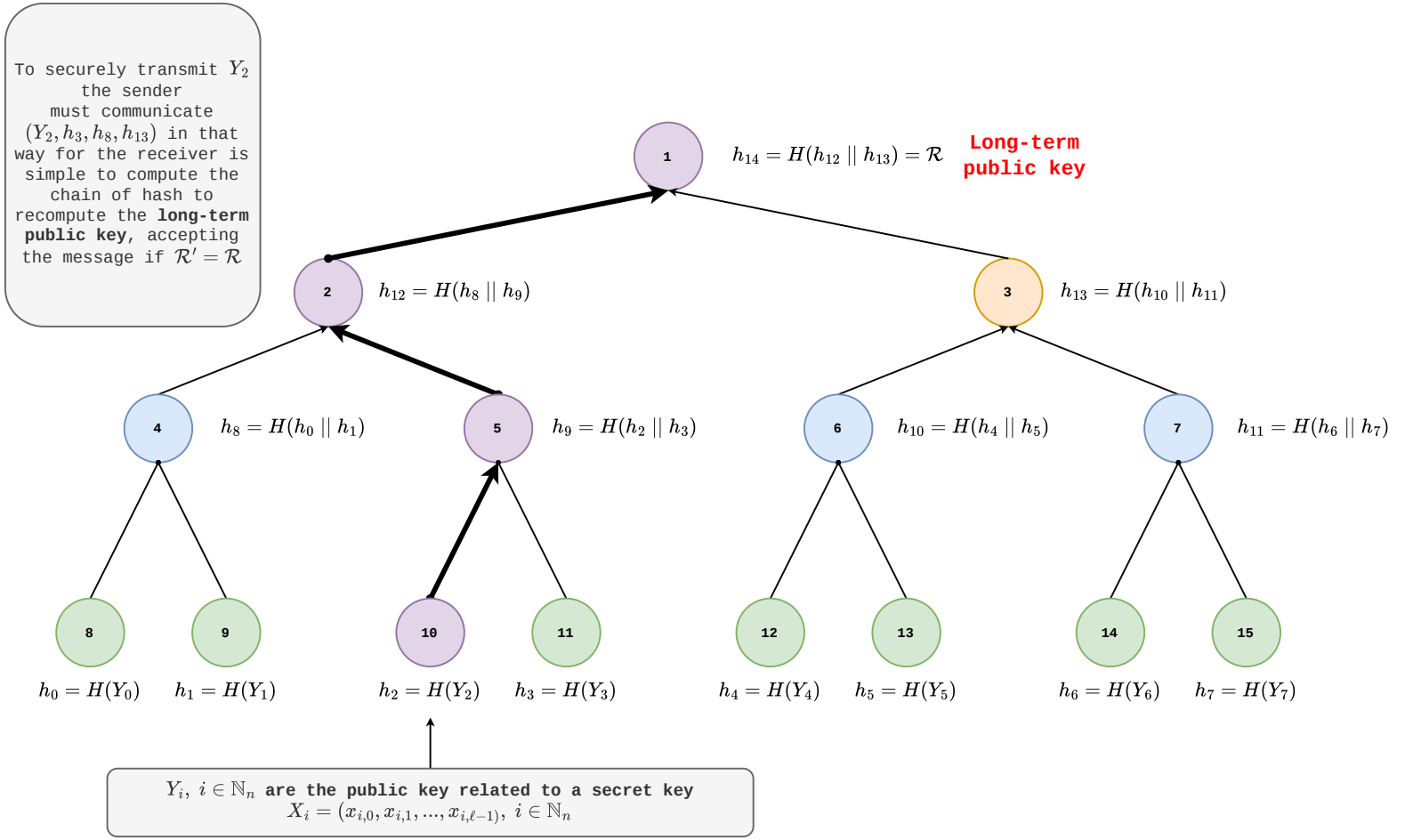
To securely transmit Y_2 the sender must communicate (Y_2, h_3, h_8, h_{13}) in that way for the receiver is simple to compute the chain of hash to recompute the **long-term public key**, accepting the message if $\mathcal{R}' = \mathcal{R}$
1
$h_{14} = H(h_{12} \,||\, h_{13}) = \mathcal{R}$
Long-term public key
2
$h_{12} = H(h_8 \,||\, h_9)$
3
$h_{13} = H(h_{10} \,||\, h_{11})$
4
$h_8 = H(h_0 \,||\, h_1)$
5
$h_9 = H(h_2 \,||\, h_3)$
6
$h_{10} = H(h_4 \,||\, h_5)$
7
$h_{11} = H(h_6 \,||\, h_7)$
8
9
10
11
12
13
14
15
$h_0 = H(Y_0)$
$h_1 = H(Y_1)$
$h_2 = H(Y_2)$
$h_3 = H(Y_3)$
$h_4 = H(Y_4)$
$h_5 = H(Y_5)$
$h_6 = H(Y_6)$
$h_7 = H(Y_7)$
$Y_i,\ i \in \mathbb{N}_n$ **are the public key related to a secret key** $X_i = (x_{i,0}, x_{i,1}, ..., x_{i,\ell-1}),\ i \in \mathbb{N}_n$